Abdulrahman Sirelkhatim

Medical Graduate · Health Data Scientist · Organizational Leader

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Professional Summary

First-generation medical graduate (MBBS, Sudan University of Science and Technology (SUST), 2026) and health data scientist with a combination of clinical training, advanced data science skills, and proven organizational leadership. Over 7 years of progressively responsible roles across medical student organizations (serving 800+ members), national and international healthcare networks, and a surgical development consortium — all while completing medical education through four major disruptions: revolution, COVID-19, faculty strike, and ongoing armed conflict.

Delivered 25 complete data analysis projects for medical researchers across 10+ specialties. Facilitated 43 sessions across 7 years (106 hours). Held top leadership positions including President of SUMSA (SUST largest medical students' association) and Deputy Director at SCSD (national surgical development consortium). Author or co-author of five research projects: two multicenter qualitative studies on mass casualty incident preparedness during the armed conflict in Sudan, a post-conflict epidemiological baseline assessment, a Grade A stroke KAP thesis, and an ongoing deep learning study on African neuroimaging data. Combines clinical neuroscience knowledge with machine learning and statistical analysis skills, targeting health data science research and graduate study in computational neuroscience.

Education

Bachelor of Medicine, Bachelor of Surgery (MBBS) — *Sudan University of Science and Technology (SUST)* · *11/2018 – 07/2026 | Khartoum, Sudan*

- **Final GPA:** 3.24/4.0 · Ranked 22nd of 127 · First Degree Honors
- **Academic progression maintained through:** 2019 Sudanese Revolution (10-month closure), COVID-19 pandemic (6-month closure), 2021 faculty strike (6 months), April 2023 armed conflict (ongoing displacement)
- **Graduation Thesis:** "Knowledge, Attitudes and Practices on Stroke Manifestations and Complications Among Sudanese Population, Khartoum 2025" — Grade A · Cross-sectional survey design · Python + SPSS analysis · Informs public health education strategy for stroke prevention in Sudan
- **Relevant Coursework:** Neuroscience (CNS/ANS), Neurology, Biostatistics, Epidemiology, Research Methods, Evidence-Based Medicine (EBM), Internal Medicine, Surgery

Certificate in Computer & Data Science — *Massachusetts Institute of Technology (MIT) Emerging Talent Program* · *11/2024 – 12/2025 | Cambridge, MA (Remote)*

- Competitive selection from international cohort (5% acceptance rate)
- Successfully completed Foundation Track (11/2024–12/2024): Python, Git/GitHub, Version Control, Discrete Mathematics, Computational Thinking, Debugging, Software Engineering Practices
- **Capstone Project:** Humanitarian Crisis Intelligence and Communication System (HCICS) — data-driven geospatial platform addressing public health information breakdown in Sudan; fused IOM Displacement Tracking Matrix (DTM) + Humanitarian OpenStreetMap Team (HOT) data; geospatial analysis mapping IDP locations against health facility availability for Ministry of Health resource allocation
- **Technology Stack:** Python (GeoPandas, Folium, Shapely, Pandas), Streamlit/Dash dashboard

Pan African Youth Leadership Program (PAYLP) — *University of Texas at Austin — US Department of State* · *07/2017 | Austin, TX, United States*

- Competitive program for outstanding African youth leaders — nominated from Access Micro-scholarship cohort, selected via interview
- **Certificates from:** University of Texas at Austin, US Department of State, Meridian International Center
- **Focus:** Leadership development, civic engagement, entrepreneurship, cultural exchange · Mini Diploma in Business Administration (UT Austin)

Access Micro-scholarship Program — *US Embassy Khartoum* · *11/2015 – 11/2017 | Omdurman, Sudan*

- 2-year program for outstanding high school students · Selected from competitive pool (95%+ GPA in first two years of secondary school)
- **Components:** English Language (600+ hours intensive instruction), Computer Skills (ICDL curriculum), Community Service (Ramadan iftar distribution, street cleaning, health awareness campaigns)
- **Scientific visits:** University of Khartoum, Presidential Villas · Graduation ceremony with US Ambassador attendance

Research Experience

Trauma Surgeons' Preparedness for Mass Casualty Incidents During the Armed Conflict in Sudan (2025) — *Multicenter Qualitative Study* · *12/2024 – 04/2025 | Sudan*

- Co-author. Multicenter qualitative study examining how trauma surgeons prepared for and managed mass casualty incidents during the armed conflict in Sudan

Doctors' Perspectives of Hospital Preparedness for Mass Casualty Incidents in Conflict-Affected Regions of Sudan (2025) — *Multicenter Qualitative Study* · *2025 | Sudan*

- Co-author. Multicenter qualitative study of doctors' perspectives on hospital-level preparedness for mass casualty incidents in conflict-affected regions

Knowledge, Attitudes and Practices on Stroke Manifestations and Complications Among Sudanese Population, Khartoum 2025 — *MBBS Graduation Thesis, SUST (Grade A)* · *05/2025 – 09/2025 | Khartoum, Sudan*

- Author. Cross-sectional survey with a structured questionnaire; descriptive and inferential analysis in Python and SPSS to identify knowledge gaps, cultural attitudes, and health-seeking behavior

Um Katti Hospital and Its Catchment Community: A Post-Conflict Epidemiological Baseline Assessment, Karari Locality, Khartoum State — *Rural Health Training Programme, SUST* · *04/2026 | Khartoum State, Sudan*

- Main author. 8-member team; household survey of 262 households (1,355 persons) with proportionate stratified systematic random sampling, 26 key informant interviews, and 8 structured-observation checklists
- Wrote the final report on disease burden, PHC programme gaps, workforce shortages, and drug stockouts for a referral facility serving about 66,000 people

Evaluating Deep Learning Brain Tumor Classification Performance on African Neuroimaging Data with Grad-CAM Explainability and Offline Mobile Deployment — *Independent Research* · *Ongoing | Remote*

- Main author. Proposal and literature review stage; planned evaluation of brain tumor classification models on African neuroimaging data with Grad-CAM explainability and offline mobile deployment for low-resource settings

Professional Experience

Organizer, Fundamentals of Nursing Course — *Sharg Alneel Hospital* · *10/2026 – 12/2026 | Khartoum, Sudan*

- Support delivery of the curriculum and facilitators' guide, keeping sessions on schedule
- Engage participants, assist facilitators, and record attendance for both participants and facilitators
- **Delivered 3 sessions to nursing staff:** Medical Ethics, Professionalism and Patients' Rights; Counselling Skills in Nursing Practice; Vital-Sign Technique, Interpretation and Monitoring
- Created a facilitator portfolio documenting 43 sessions and 106 hours of training delivered across 9 universities and national programs (2020–2026)

Freelance Health Data Analyst — *Self-Employed* · *10/2025 – Present | Khartoum, Sudan*

- Delivered 25 complete data analysis projects for MD, MBBS, Clinical Master, and Doctoral researchers across 10+ medical specialties including oncology, infectious disease, psychiatry, ophthalmology, maternal health, and public health
- **Full-service statistical support:** study design consultation, data cleaning and wrangling, exploratory data analysis, descriptive and inferential statistics, multi-variable regression modeling, survival analysis, diagnostic test evaluation (sensitivity, specificity, ROC curves, predictive values), longitudinal data analysis
- **Deliverables per project:** comprehensive statistical report with methodology explanation, publication-ready visualizations (charts, graphs, heatmaps), PowerPoint thesis defense presentation, interpretation of results with clinical significance context
- **Tools:** Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib, seaborn), SPSS, Excel
- Earned $350+ in first 6 weeks while supporting family during armed conflict and displacement
- GitHub repositories documenting the projects (portfolio update in progress)
- 11 thesis defense presentations prepared for client defense sessions; all clients successfully defended

Deputy Director, Resources & Excellence Management — *Sudanese Consortium for Surgical Development (SCSD)* · *08/2024 – 05/2026 | Remote*

- Supported department director in managing human resources strategy, recruitment, and organizational development for pan-Sudanese surgical advancement consortium with signed MoU with Sudan Federal Ministry of Health
- **Recruitment leadership:** opened 5+ calls, reviewed 80+ applications total across 4 departments + COO Assistant position, conducted 20-30 interviews, managed candidate screening and onboarding processes
- Designed and implemented continuous professional development programs including organizational training framework and capacity building initiatives
- Defined Key Performance Indicators (KPIs) for all organizational positions from Chief Visionary Officer to entry-level officers
- Developed annual monitoring and evaluation framework for all departments
- Drafted partnership guidelines, operational frameworks, and strategic planning documents; managed Board of Advisors engagement
- Acted as department director during 5-week absence; attended executive board meetings; facilitated COO Assistant recruitment
- Managed monthly recognition emails, performance feedback systems, conflict resolution with inactive members across all departments

Leadership Experience

Member — *Sudan University Medical Students Association (SUMSA) — Advisory Board* · *08/2026 | Khartoum, Sudan · 6-member board*

- Invited by SUMSA's current leadership to join the organization's first-ever Advisory Board (6 members with equal standing); board formation decided August 2026, evidence that governance systems built during earlier terms continue to evolve independently of the original leadership

President — *Sudan University Medical Students Association (SUMSA)* · *03/2025 – 12/2025 | Khartoum, Sudan · 800+ students*

- Led executive committee of 8 members serving 800+ medical students across 5 academic years — highest executive position in the organization
- **Organizational rescue:** rebuilt SUMSA from near-collapse by personally recruiting executive committee members, filling 2 vacant secretariat positions, and restoring full operational capacity within 2 months

- **Designed the SUMSA M&E system — features:** term management, member database, monthly/quarterly/annual evaluations, weighted scoring (5-pillar system), automated award generation, progress visualization charts
- MedXplore 2025 (Co-Coordinator): designed and delivered 7-day online orientation program for 465 students (56% of 833 total student body) across all 7 batches; aligned with CanMEDS Physician Competency Framework; covered academic success, clinical/non-clinical career pathways, research, leadership, communication, time management, CV building, mentorship
- SUMSA Professional Development Program (SPDP) (Lead Organizer): launched 16-session professional capacity building program attended by 216 members across 6 weeks; topics included organizational management, leadership, strategic planning, financial management, marketing, event management, accountability
- **Highlights ceremony:** organized Oscar-style awards ceremony acknowledging executive committee achievements and team contributions
- **Constitutional reform:** led 4-year review process through executive committee vote, council approval, and general assembly ratification (400+ voters); all proposed amendments passed
- **Designed comprehensive credit hours system:** base hours by position + contribution multiplier tied to M&E evaluation scores; governing bylaws + certificates issued
- Formed and headed independent evaluation committee with bylaws governing formation, job descriptions, evaluation process, report format, deadlines, and objection/appeal processes
- Headed Elections Committee for 2026 term: managed 400+ voter election using Google Forms with identity verification; built App Script to automate batch-by-batch vote tabulation; announced results within 24 hours; oversaw second call for Secretary General

Secretary General (De Facto President) — *Sudan University Medical Students Association (SUMSA)* · *02/2023 – 12/2023 | Khartoum, Sudan | 7-member executive committee*

- Directed all association operations as de facto president (no president elected that year) while fulfilling full Secretary General duties
- **DIGITAL ARCHIVE:** established 14,000+ document archive from scratch over 3 months — collected from 38+ email accounts, telegram channel, Facebook page, individual drives; classified into 13 categories; created naming convention [Term][Name]; uploaded to dedicated archive email; drafted archive bylaws and formed archiving committee for future terms
- Organized 36 meetings total (16 council, 5 executive committee, 15 executive board); drafted all agendas, recorded all minutes within 48 hours, published outcomes to general assembly
- Submitted 8 monthly reports to General Council documenting activities, challenges, and strategic recommendations
- **Led fundraising committee coordination:** raised 12,000,000+ SDG (~$5,000 equivalent) supporting 98+ war-affected students; prevented academic dropout for dozens of students
- Issued 230+ certificates of experience across 3 organizational terms using credit hours system
- **Constitutional compliance:** achieved 100% adherence to bylaws; interpreted ambiguous provisions; formed investigation committee for violations; updated constitution post-General Assembly
- **Headed reform committee:** evaluated 4 years of organizational history, identified 3 systemic defects (documentation gaps, lack of institutional memory, no feedback mechanism), proposed solutions (archive, M&E system, credit hours — all implemented as President)
- Represented executive board at White Nile medical mission organized by Public Health Secretary
- Crisis adaptation (April 2023 armed conflict): pivoted entire organization to academic stability support; mobilized fundraising; maintained operations with 0 days of unplanned downtime

Head of Elections Committee — *Sudan University Medical Students Association (SUMSA)* · *10/2025 – 12/2025 | Khartoum, Sudan*

- Managed all electoral processes for 2026 term: candidate nominations, eligibility verification against constitution, voting procedures, results verification
- Built Google Forms voting system with 4-section identity verification; created Google Sheets App Script to filter and tabulate 400+ votes by batch; announced results within 24 hours
- Oversaw constitutional amendment vote alongside elections; all amendments approved

Fundraising Committee Head — *Sudan University Medical Students Association (SUMSA)* · *06/2025 – 09/2025 | Khartoum, Sudan | 8-member committee*

- Led 8-member committee supporting students financially affected by April 2023 armed conflict
- Since 06/2025: served 15+ students with 1,200,000+ SDG in direct support
- Overall committee (Aug 2023–present): 98+ students supported, 12,000,000+ SDG raised

External Affairs Secretary — *Sudan University Medical Students Association (SUMSA)* · *06/2022 – 01/2023 | Khartoum, Sudan*

- Drafted and managed 371 letters to government entities, NGOs, private sector, and international organizations — single-day record: 59 letters in one day for medical mission funding deadline
- Secured $14,000+ in financial and in-kind support for White Nile Medical Mission and other activities
- Represented association at external events, conferences, and stakeholder meetings; signed collaboration contracts
- **Initiated exchange program groundwork:** secured dean's approval, began Khartoum Teaching Hospital partnership negotiations (program continued by successors; currently in guide preparation phase)

Students Council Member (Board Member) — *Sudan University Medical Students Association (SUMSA)* · *2019–2021, 2022, 2023, 2025 | Khartoum, Sudan*

- **Participated in council governance:** approved internal regulations, strategic plans, periodic performance reports, and bylaws
- Oversaw and monitored executive committee activities; guided decisions on organizational membership in local, regional, and international networks

- Participated in 7 interview committees (3 for SUMSA: 2 as SecGen, 1 as expert; 4 for MedSIN-SUMSA as VPE)

Batch Coordinator — *SUST College of Medicine* · *05/2022 – 07/2026 | Khartoum, Sudan*
- Served as primary liaison between 4-member coordination team and entire academic batch (130+ students)
- Facilitated communication between students, faculty, and administration
- Coordinated batch-wide academic support, activities, and addressing student concerns throughout final years of medical school

IFMSA / MedSIN-Sudan (National and International)

MedSIN-Sudan is Sudan's representative body in IFMSA (The International Federation of Medical Students' Associations). MedSIN-SUMSA is SUMSA's local representative office. SCOPE (Standing Committee on Professional Exchange) coordinates international medical student exchange programs and global health education.

General Assistant — SCOPE National — *MedSIN-Sudan* · *01/2023 – 12/2023 | Khartoum, Sudan*
- Organized first-ever Online General Assembly post-conflict: managed sessions materials, timetable, facilitator coordination, member engagement in virtual environment — achieved one of highest satisfaction rates among all standing committees
- Prepared meeting invitations, agendas, minutes; updated committee bylaws; fulfilled national officer duties during absences
- Represented MedSIN-Sudan SCOPE at IFMSA international sessions and delegation meetings; participated in exchange fair

Capacity Building Development Assistant — SCOPE National — *MedSIN-Sudan* · *07/2021 – 12/2021 | Khartoum, Sudan*
- Organized first national SCOPE activity in 5 years: 3-day Global Health and Universal Health Coverage workshop — managed full logistics from proposal to certificates (venue, calls, selection, facilitators, materials, budget, financial report)
- Founded TNEF (Training National Exchange Facilitators) workshop: designed curriculum, selection criteria, graduation requirements, organizational governance framework — introduced at first National Pre-General Assembly Workshop; increased certified facilitators from 7 to 20+; now established as official SCOPE Sudan training program
- Drafted first-ever SCOPE Sudan Capacity Building Guidelines — included TNEF governance, session standards, and facilitator requirements
- Created national trainers (7) and facilitators (12) database; supervised 11 university local activities; assisted NEO with SCOPE General Assembly session preparation (timetable, guiding booklet, pre/post evaluations, logistics)
- Named "Assistant of the Term" — 100% of planned objectives achieved

Vice President for External Affairs — *MedSIN-SUMSA* · *06/2022 – 01/2023 | Khartoum, Sudan*
- Attended national LMO (Local Member Organization) meetings with MedSIN-Sudan Executive Board representing SUMSA
- Prepared MedSIN-SUMSA delegations for General Assemblies: opened calls, conducted interviews, selected representatives
- Member of MedSIN-SUMSA 3-year strategy drafting committee and projects regulations expert committee

Local Exchange Officer (LEO) — SCOPE — *MedSIN-SUMSA* · *06/2021 – 05/2022 | Khartoum, Sudan*
- Organized and facilitated training sessions; attended Team of Officials meetings; submitted monthly committee reports and plans
- Attended MedSIN-Sudan General Assembly as SUMSA representative in SCOPE sessions
- Advanced exchange program activation requirements: secured dean's approval, assisted successor with hospital partnership negotiations

Development Assistant — SCOPE — *MedSIN-SUMSA* · *06/2020 – 01/2021 | Khartoum, Sudan*
- Supported SCOPE office establishment during founding phase; co-organized 2 training sessions including international IFMSA facilitator coordination
- Attended Local Officers' Camp organized by MedSIN-Sudan for skills development in standing committee operations

SCOPE Initiatives Reform — Small Working Group — *IFMSA International Team* · *11/2023 – 06/2024 | Khartoum, Sudan*
- Worked under supervision of SCOPE General Assistant on international team; researched and drafted sections of SCOPE Initiatives manual (trilateral exchanges, dental exchanges, extended-duration exchanges) distributed to SCOPE members worldwide

Certified Facilitator — IFMSA — *PRET Training, Egypt* · *05/2022 | Khartoum, Sudan*
- Earned certification at Professional and Research Exchange Training (PRET) — IFMSA international training on exchange program management and facilitation methodology
- **Total facilitation record:** 43 sessions, 106 hours across 7 years (2019–2026) — topics: exchange programs, global health, UHC, leadership, teamwork, intercultural learning, professional skills

SCOPE Plenary Secretary — *MedSIN-Sudan General Assembly* · *2022 | Khartoum, Sudan*
- **Managed SCOPE plenary sessions:** presentation, attendance, minutes, voting oversight

Field & Clinical Experience

Group Head (Group B) — *SUMSA Medical Mission — Garri Area, Khartoum State* · *06/2026 | Field Deployment*
- Led Group B field deployment (32 students + 8 lab members + pharmacist + security) serving 600–650 patients across medical day operations in one of Khartoum's most war-affected areas
- **Responsibilities:** morning check-in, tag management, overseeing 6 offices (clinics, pharmacy, lab, research, health education, environmental health), anticipating and resolving real-time operational problems, daily team debrief, end-of-day logistics

Vice Head, Organizing Committee — *SUMSA Medical Mission — Garri* · *12/2025 – 09/2026*
- **Pre-mission:** prepared working folder, sent invitations, wrote meeting minutes, assisted head with overseeing office heads, prepared all planning templates, prepared full mission timeline from committee formation to report submission

- **Post-mission:** assisted head in reviewing all office reports and compiling final mission report for Public Health Secretary

Executive Board Representative — *White Nile State Medical Mission* · *02/2023*
- Represented SUMSA executive board at mission organized by Public Health Secretary; oversaw operations and resolved escalated issues

MBBS Clinical Training — *Sudan University of Science and Technology* · *2022 – 2026 | Multiple Teaching Hospitals*
- **Full clinical rotations across:** Internal Medicine, General Surgery, OB/GYN, Pediatrics, Ophthalmology, ENT, Orthopedics, Neurosurgery, Pediatrics Surgery and Urology
- Coordinated OB/GYN clinical rounds as batch coordinator for 130+ students
- Completed 132-page rural residency report

Rural Health Trainee — *Rural Health Training Programme, Department of Community Medicine, SUST* · *04/2026 | Um Katti Hospital, Karari Locality, Khartoum State, Sudan*
- Conducted a post-conflict epidemiological baseline assessment as part of an 8-member student team for Um Katti Hospital, sole referral facility for ~66,000 people across 19 villages — the first comprehensive post-conflict assessment of the site
- Co-designed and administered a household survey of 262 households (1,355 persons) using proportionate stratified systematic random sampling (Finite Population Correction formula, 95% CI, ±5% margin of error)
- Conducted 26 key informant interviews with 21 hospital personnel using 17 self-developed bilingual (Arabic/English) interview guides
- Built and applied 8 standardized structured-observation checklists spanning infrastructure, environmental health, drug availability, EPI, nutrition, IMCI, and reproductive health programme performance
- Main author of the team's final English-language report, quantifying disease burden, PHC programme gaps, workforce shortages, and pharmaceutical stockouts for submission to the Department of Community Medicine

Selected Projects

Humanitarian Crisis Intelligence & Communication System (HCICS) — *MIT Emerging Talent Capstone · Python · GeoPandas · Folium · Streamlit* · *10/2025 – 12/2025*
- **Data aggregation engine:** collected, cleaned, and fused humanitarian data from IOM DTM API and Humanitarian OpenStreetMap (HOT)
- **Geospatial correlation:** mapped IDP locations against health facility availability; identified underserved populations
- **Dashboard:** interactive crisis visualization for Ministry of Health resource allocation decision-making

Knowledge, Attitudes and Practices on Stroke Manifestations and Complications — MBBS Graduation Thesis — *SUST · Python · SPSS · Grade A* · *2025*
- **Primary research:** cross-sectional survey design, structured questionnaire, Sudanese population sample
- **Analysis:** descriptive and inferential statistics; knowledge gaps, cultural attitudes, health-seeking behavior patterns
- **Significance:** informs public health education strategy for stroke prevention in Sudan; directly relevant to Sudan's near-absent stroke awareness infrastructure

SpaceX Falcon 9 Landing Prediction — *IBM Data Science Capstone · Python · scikit-learn · Plotly* · *2024*
- Predicted rocket landing success using machine learning (classification); built interactive dashboard visualization

25 Medical Research Data Analysis Projects (Portfolio) — *Freelance · Python · SPSS* · *2024–2026*
- **Specialties covered:** infectious disease (malaria, dengue, TB, cholera, diarrheal disease), psychiatry (depression, academic stress), ophthalmology, maternal/reproductive health, pharmacy (QSAR/molecular docking), public health, medical education research
- Full project list and GitHub repositories available on request

Technical Skills

Programming & Data Science
- **Python (advanced):** pandas, NumPy, scikit-learn, SciPy, statsmodels, matplotlib, seaborn, plotly, GeoPandas, folium, shapely
- **Statistical Software:** SPSS (advanced — descriptive, inferential, regression, survival, ROC), Excel (advanced — functions, pivot tables, statistical analysis)
- **Machine Learning:** supervised learning (regression, classification), unsupervised learning (clustering, dimensionality reduction), model evaluation, cross-validation, ROC analysis
- **SQL:** database querying, joins, aggregations, data manipulation
- **Environments:** Jupyter Notebooks, VS Code, Google Colab
- **Version Control:** Git, GitHub (repositories, branching, commits, documentation)
- **Scripting:** Google Apps Script (automated vote tabulation for 400+ voters)

Productivity & Collaboration
- **Microsoft Office Suite:** Word, Excel, PowerPoint (advanced)
- **Google Workspace:** Docs, Sheets, Slides, Forms, Drive (advanced — including Apps Script)
- **Project Management:** Notion, Asana (functional knowledge)
- **Database & Archive Management:** digital classification systems, document management, naming conventions, access protocols

Clinical & Medical
- **Clinical skills:** full clinical rotations across 10 specialties

- **Research:** study design, survey methodology, statistical analysis, scientific writing, thesis supervision support

Leadership & Soft Skills

- **Executive Leadership & Operations:** Crisis Leadership & Operations, Delegation & Succession Planning, Systems Design & Process Optimization
- **People & Resource Management:** Cross-Functional Team Building, Recruitment, Financial Stewardship under Scarcity
- **Communication & Engagement:** Stakeholder Management, Written and Verbal Communication
- **Adaptability & Mindset:** Resilience, Comfort with Ambiguity, Growth Mindset, Adaptability

Certifications & Professional Development

Technical Certifications

- IBM Data Science Professional Certificate · Coursera · 02/2024 · 10-course specialization covering full data science pipeline: Python, SQL, data visualization, machine learning, capstone project
- DeepLearning.AI Mathematics for Machine Learning and Data Science Specialization · Coursera · 06/2025 · 3 courses: Linear Algebra, Calculus, Probability & Statistics for ML
- Google Project Management Professional Certificate · Coursera · 08/2024 · 6 courses: project planning, execution, Agile, Scrum, risk management, stakeholder communication
- IBM Data Science Fundamentals with Python and SQL · Coursera · 10/2024 · 5 courses: descriptive/inferential statistics, hypothesis testing, regression, ANOVA
- MITx 6.00.1x Introduction to Computer Science and Programming Using Python · edX · 04/2025
- MITx 6.00.2x Introduction to Computational Thinking and Data Science · edX · 05/2025
- MIT Emerging Talent Program — Foundation Track · 11–12/2024 · Python, Git/GitHub, Discrete Mathematics, Computational Thinking, Debugging

Medical & Healthcare Certifications

- Certified Facilitator · International Federation of Medical Students' Associations (IFMSA) · 05/2022 · 90+ hours of facilitation training — PRET, Egypt
- IFMSA Online Global Health Advocacy (OGHA) Workshop · International Federation of Medical Students' Associations · 09–10/2022 · six-session program: UN and WHO, 2030 Agenda, global health diplomacy, maternal health and UHC, advocacy and stakeholder mapping, youth inclusion in decision-making, IFMSA policies, advocacy simulations
- Research Methodology Workshop · MedStat · 01/2021

Leadership & International Programs

- Pan African Youth Leadership Program · University of Texas at Austin · 07/2017 · US Department of State + Meridian International Center · Mini Diploma in Business Administration
- Access Micro-scholarship Program · US Embassy Khartoum · 11/2015 – 11/2017 · 600+ hours English instruction, ICDL computer skills, community service
- Member of MedSIN-Sudan Delegation to IFMSA March Meeting 2024 – Online

Languages

Arabic — Native (Mother Tongue)

English — Fluent (C1 level, Duolingo English Test 140) · 600+ hours intensive instruction (US Embassy Access Program 2015–2017) · Professional proficiency in academic writing, technical documentation, public speaking, and presentations · Self-directed learning through academic reading, online courses, and professional communication since 2015

References

Available upon request · References available from SUMSA, SCSD, MedSIN-Sudan, MIT Emerging Talent Program, and freelance research clients